

Euiseok Han

✉ hanes1207@kaist.ac.kr

🐙 hanes1207.github.io

🌐 linkedin.com/in/hanes1207

RESEARCH INTERESTS

- Uncertainty-Aware Decision Making for Autonomous Systems: Reliability-Aware Multimodal Perception, Uncertainty-Aware State Estimation, Uncertainty-Aware Robot Policies, 3D Scene Understanding

EDUCATION

- Korea Advanced Institute of Science and Technology, Daejeon** Mar. 2020 – Aug. 2027 (Expected)
B.S. in Electrical Engineering and Computer Science (Double Major) GPA: 4.02/4.3

– *Coursework*: Machine Learning for 3D Data, Kalman Filtering (graduate), Mathematical Physics

- University of Texas at Austin** Aug. 2025 – Dec. 2025
Exchange Program GPA: 4.0/4.0

PUBLICATIONS

- [1] **SenseFuse: Label-Free Fusion of Image and Shape Encoders for Open-Vocabulary 3D Instance Segmentation** Under review
Euiseok Han, Tri Ton, Hwanhee Kim, Seungyeon Ryu, Chang D. Yoo
ICRA 2027 – IEEE International Conference on Robotics and Automation [arXiv:2609.20475] [code]
- [2] **Reference-Filter-Driven Transition Probabilities for IMM-Based Satellite Maneuver Detection** Preprint
Euiseok Han, Sangheon Choi, Seung-Hyun Kong
arXiv preprint [arXiv:2610.11242] [interactive demo]

RESEARCH EXPERIENCE

- Artificial Intelligence and Machine Learning Lab, KAIST** Jan. 2025 – Feb. 2025, Jan. 2026 – Present
Undergraduate Researcher / Advisor: Prof. Chang D. Yoo Daejeon, KR

- URP: *From Point Cloud to Actions: 3D Structural Keypoint-Conditioned Hierarchical Robot Policy Learning*
- Revealed that frozen 2D image and 3D shape encoders fail differently and are weakly correlated (Spearman ρ 0.08–0.23, 5.9% error overlap), versus 0.68 and 41.4% for a 2D–2D pair [1]
- Devised a 2D-3D fusion framework that enhances mask labeling through sensitivity maximization
- Proposed a closed-form, label-free scene-level weight calibration method operating in 3.5 ms, achieving a median gain of +7.1 points and recovering 93% of the oracle-weight gain

- NC Lab, KAIST** Mar. 2025 – Aug. 2025
Undergraduate Researcher / Advisor: Prof. Junehwa Song Daejeon, KR

- *Socially-appropriate Use of Noise-making Everyday Appliances in High-density Urban Housing*
- Participated in experiment design and conducted a preliminary experiment using Zigbee modules

- Control and Intelligent Systems Lab, KAIST** Jul. 2024 – Aug. 2024
Undergraduate Researcher / Advisor: Prof. Heejin Ahn Daejeon, KR

- Reproduced Constrained Policy Optimization (ICML 2017) baselines in MuJoCo from the official code (safe RL)

SELECTED PROJECTS

•LLM-Guided Autonomous Robots

Mar. 2026 – Jun. 2026

EE40078 Introduction to Multi-disciplinary Robotics, School of Electrical Engineering

- LLM-guided autonomous mobile delivery robot; RTAB-Map localization, path planning, and object pick-up

•Hamiltonian Invariant-Driven Mode-Transition IMM for Satellite Maneuver Detection

Mar. 2026 – Present

MO50125 The Principles and Applications of the Kalman Filter, Graduate School of Mobility

- IMM transition probabilities driven by a measurement-side CUSUM on the J_2 -Hamiltonian residual
- Developed into RD-IMM, driving the transitions from the NIS of a reference coast filter outside the IMM [2]

•PintOS-KAIST

Sep. 2024 – Dec. 2024

CS330 Operating Systems, School of Computing

- Implemented kernel modules for process scheduling, syscall, memory management of an educational x86-64 OS

HONORS AND AWARDS

•Korea-U.S. Advanced Industries Youth Exchange Scholarship

Aug. 2025

\$9,000 for advanced-industry exchange study in the U.S.

OTHER EXPERIENCE

•Republic of Korea Air Force (mandatory military service)

Apr. 2022 – Jan. 2024

Information System Operator, Staff Sergeant, KAMD Operations Center

Osan Air Base, KR

- Operated real-time data processing servers for the Korea Air and Missile Defense center; 24/7 availability

TECHNICAL SKILLS AND INTERESTS

Programming Languages: C, C++, Python

Libraries: ROS, NumPy, PyTorch

Miscellaneous tools: MATLAB, L^AT_EX, Docker

Extracurricular: STROKE (Tennis Club)